

Pleural Manometry in Pneumothorax

Evaluating Tension Physiology and Predicting Outcomes



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BACKGROUND: Tension pneumothorax (TP) traditionally is defined pleural pressure (P_{pl}) exceeding atmospheric pressure, which reduces venous return and leads to hemodynamic collapse. However, P_{pl} has yet to be measured directly in TP because of technical limitations, nor has it been studied in predicting outcomes for pneumothorax.

RESEARCH QUESTION: Do increased pleural pressures increase the risk of TP and are they associated with pleurodesis or death?

STUDY DESIGN AND METHODS: In this multicenter, prospective, observational study, participants with newly diagnosed pneumothorax (tension and nontension) were recruited. P_{pl} throughout the respiratory cycle was measured at the time of chest thoracostomy, and patients were followed up for outcomes of pleurodesis and death during hospitalization.

RESULTS: At 5 centers, 37 patients with a pneumothorax (9 tension, 28 nontension) were enrolled. Seventy-three percent of recruited patients had an iatrogenic pneumothorax. Both increased inspiratory P_{pl} (OR, 1.40; 95% CI, 1.10-1.76) and expiratory P_{pl} (OR, 1.22; 95% CI, 1.05-1.42) were associated with increased risk of tension. A subset of patients without tension were found to have supraatmospheric P_{pl} . Lower inspiratory P_{pl} also was associated with an increased risk of pleurodesis (OR, 0.66; 95% CI, 0.45-0.96).

INTERPRETATION: Our results show that measurement of P_{pl} in TP is feasible. Elevated P_{pl} alone may not dictate the development of TP. In this small study, with most participants demonstrating iatrogenic pneumothorax, an association between P_{pl} and need for pleurodesis was found that may be a useful predictive tool to guide earlier definitive management. These findings should be explored further in other causes of pneumothorax as well.

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ABBREVIATIONS: CVC = central venous catheter; EP_{pl} = end-expiratory pleural pressure; IP_{pl} = end-inspiratory pleural pressure; MP_{pl} = mean pleural pressure; P_{pl} = pleural pressure; TP = tension pneumothorax

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Take-Home Points

Research Question: Are increased pleural pressures associated with tension pneumothorax (TP) and can they be used to predict clinical outcomes?

Results: In 37 patients with pneumothorax, higher pleural pressures were associated with statistically significant increased odds of tension, whereas lower inspiratory pleural pressures were associated weakly with the need for pleurodesis.

Interpretation: The results of this study support the role of increased pleural pressures in the development of TP and provide evidence for the potential predictive usefulness of pleural manometry for pleurodesis.

Pneumothorax is the presence of air within the pleural space. Tension pneumothorax (TP) is a life-threatening type of pneumothorax that historically has been defined as pleural pressure (P_{pl}) exceeding atmospheric pressure for most of the respiratory cycle, which impedes venous return and, in turn, drastically reduces cardiac output.¹⁻³ Alternatively, it has been suggested that most patients not receiving positive pressure ventilation primarily experience respiratory arrest because of hypoxia, as opposed to a primary cardiovascular collapse.⁴ TP and select cases of nontension pneumothorax are treated with chest thoracostomy for decompression until air ceases to accumulate within the pleural space. In patients with persistent air leak, more invasive interventions like thoracoscopy and pleurodesis can be performed.⁵ Currently, we have no data aiding prediction of which patients will need these advanced therapies at the time of diagnosis. Furthermore, to our knowledge, the very definition of

TP has never been confirmed because P_{pl} in patients with TP has never been measured directly. Herein, we aimed to understand better the physiologic characteristics of TP via direct measurement of pleural pressure via manometry.

Although limited data exist evaluating the role of pleural manometry in pneumothorax, it has been described in the evaluation of pleural effusion.⁶ Measurement of P_{pl} during thoracentesis facilitates safe drainage of volumes of > 1 L, prevents development of excess negative P_{pl} , diagnoses a nonexpandable lung, and predicts pleurodesis success.⁷⁻⁹ However, this has not translated to improved clinical outcomes in randomized control trials.¹⁰ In pneumothorax, routine use of pleural manometry has yet to be adopted because of the historical difficulty in obtaining measurements. However, the feasibility of pleural manometry in pneumothorax now has been demonstrated using a portable digital manometer, and a potential association between P_{pl} and prolonged tube thoracostomy drainage has been demonstrated.^{9,11,12} Still, data in patients with TP are lacking because of the increased urgency of pleural evacuation that precludes precise measurements.

Here, to our knowledge, we present among the first descriptions of directly measured P_{pl} in TP and demonstrate the feasibility of using a digital manometer in these patients. In a multicenter study, we compared P_{pl} in TP and nontension pneumothorax and evaluated the association of P_{pl} and patient characteristics with need for pleurodesis and death. We hypothesize that elevated pleural pressures correlate with tension physiologic findings, need for pleurodesis, and mortality and aim to shed further light on the pathophysiologic characteristics of TP.

Study Design and Methods

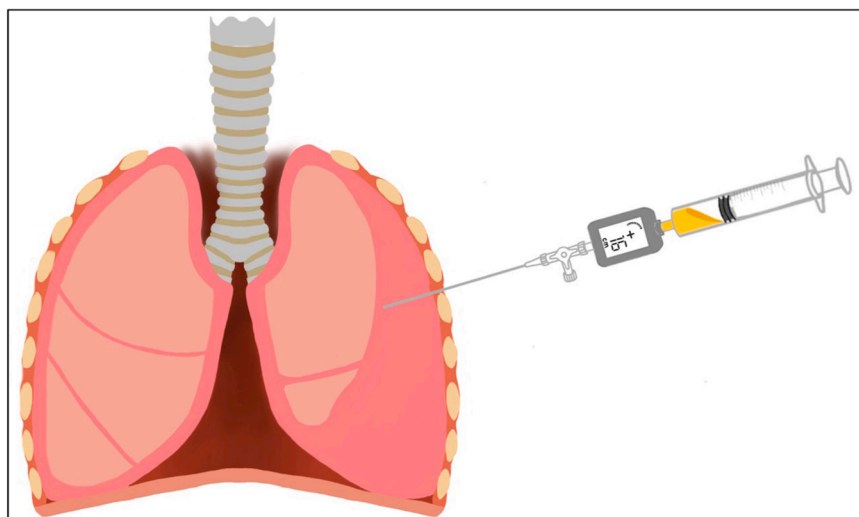
This is a multicenter, prospective, observational study enrolling patients with radiographic evidence of a new pneumothorax and planned tube thoracostomy. Patients with bilateral pneumothorax or pregnancy were excluded. Vital signs were recorded immediately before and after tube thoracostomy. Informed consent was obtained from each patient; the study was performed with Johns Hopkins School of Medicine Institutional Review Board (Identifier: IRB00256185) approval and was registered on [ClinicalTrials.gov](https://clinicaltrials.gov) (Identifier: NCT04630301). This study relied on convenience

sampling because wider sampling was limited by factors relating to the COVID-19 pandemic. As such, no formal power analysis was performed a priori.

Pleural Pressure Measurement

A digital manometer (Medline Industries, Inc.) was placed in-line with an introducer needle and syringe (Fig 1). Immediately on insertion into the pleural space, pleural pressure readings throughout the respiratory cycle were video recorded to ensure accurate data capture. The peak end-inspiratory and end-expiratory pleural pressures observed during the first 3 respiratory cycles

Figure 1 – Schematic showing pleural pressure measurement using an in-line digital manometer as performed in this study.



were identified and recorded. All patients then underwent standard tube thoracostomy.

Outcome Measurement

The primary outcomes measured were end-inspiratory and end-expiratory pleural pressures and the presence of radiographic or physiologic TP. TP was defined as a systolic BP of < 90 mm Hg, mean arterial pressure of < 65 mm Hg, initiation of vasopressors, or radiographic evidence of midline shift with resolution after tube thoracostomy placement. Pneumothorax size was measured apically and at the hilum via standard chest radiography. Participants were observed during hospitalization until discharge or death. Secondary outcomes of interest were all-cause mortality during hospitalization, pneumothorax characteristics (ie, iatrogenic, primary or secondary spontaneous), and need for additional pleural procedures (ie, surgical or chemical pleurodesis, endobronchial valve placement, additional drain placement). Patients with documented history of interstitial lung disease, COPD, lung cancer, or cystic fibrosis were considered to have structural lung disease.

Statistical Analysis

Continuous variables for the entire study cohort were reported with descriptive statistics including mean and SD. Mean pleural pressure (MP_{pl}) was calculated as the simple average of inspiratory pleural pressure (IP_{pl}) and expiratory pleural pressure (EP_{pl}). Unvari-

able logistic regression models were created to measure the associations between pleural pressures and the development of TP, as well as additional subgroup analyses for radiographically defined tension (midline shift on radiograph) and hemodynamically defined tension (initiation of vasopressors or systemic hypotension). Similar models were created to evaluate the association between pleural pressures and secondary outcomes of the need for pleurodesis and in-hospital mortality. All models were tested for adequate fit based on Hosmer-Lemeshow tests. For analytic purposes, MP_{pl} was treated as a binary variable (> 0 cm H_2O or ≤ 0 cm H_2O).

Four patients in this study were receiving positive pressure ventilation at the time of pleural pressure measurement. Because the presence of positive pressure ventilation inherently affects pleural physiologic features, it was important to examine the impact these patients had on the results of our analyses. Therefore, we performed a sensitivity analysis by recreating the logistic regression models while excluding the 4 patients receiving positive pressure ventilation. The small overall sample size and low rate of tension within the cohort precluded the ability to build multivariable regression models. All models were checked for fit using Hosmer-Lemeshow tests. For all tests, a P value of $< .05$ was deemed statistically significant. All statistical analyses were performed with Stata version 18.0 software (StataCorp).

Results

Study Population

Across 5 US-based health centers, 37 participants were enrolled between February 2021 and November 2023. The enrolled participants consisted of 21 male individuals with a mean age of 64.5 years, mean BMI of 25.0 mg/m², and mean duration of smoking of 27.1 pack-years (Table 1). Nine patients were identified to have a TP; 8 patients displayed mediastinal shift on radiography and 2 patients experienced hemodynamic

TABLE 1 Summary of Study Participant Demographics and Pneumothorax Characteristics (N = 37)

Characteristics	Data
Participant	
Male sex	21 (56.8)
Age, y	64.5 (14.3)
Structural lung disease	24 (64.9)
Pneumothorax history	4 (10.8)
BMI, kg/m ²	25.0 (6.6)
Smoking duration, pack-years	27.1 (19.8)
Mechanical ventilation	4 (11.1)
Pneumothorax	
Right side	22 (59.5)
Iatrogenic	27 (73.0)
Bronchoscopy	19 (51.35)
Bronchial stent placement	1 (2.7)
Endobronchial valve	3 (8.1)
Transbronchial biopsy	5 (13.5)
Intubation	6 (16.2)
Thoracentesis	4 (10.8)
CVC insertion	1 (2.7)
PPM insertion	1 (2.7)
Primary spontaneous	5 (13.5)
Secondary spontaneous	4 (10.8)
Traumatic	1 (2.7)
Size at apex, cm	5.5 (2.8)
Size at hilum, cm	2.3 (1.6)
Tension	9 (24.3)
Radiographic	8 (21.6)
Hemodynamic	2 (5.4)
Nontension	28 (75.7)
Inspiratory P _{pl} , cm H ₂ O	0.5 (6.3)
Expiratory P _{pl} , cm H ₂ O	6.0 (7.1)
Mean P _{pl} , cm H ₂ O	3.2 (5.8)

Data are presented as No. (%) or mean (SD). CVC = central venous catheter; P_{pl} = pleural pressure; PPM = permanent pacemaker.

compromise. None of the patients with tension had a prior history of pneumothorax, whereas 4 patients in the nontension group had a documented history of pneumothorax. Structural lung disease was documented in 24 patients. A total of 27 participants (73%) had iatrogenic causes. Of the iatrogenic pneumothoraxes, within the 48 hours preceding diagnosis, 19 patients underwent bronchoscopy, 6 patients had been intubated endotracheally, 4 patients underwent thoracentesis, 1 patient had a central venous catheter placed, and 1 patient underwent permanent pacemaker placement. Of the 6 patients recently intubated, 4 patients continued to receive mechanical ventilation at the time of diagnosis. Mean pneumothorax size was 5.5 cm at the apex and 2.2 cm at the hilum. Figure 2 demonstrates scatterplots of observed end-inspiratory and end-expiratory pleural pressures for each study participant, by tension status, demonstrating that all patients with TP had both positive IP_{pl} and EP_{pl}. However, many patients without tension also had both positive IP_{pl} and EP_{pl}. Over the entire cohort, mean IP_{pl} was 0.5 cm H₂O, mean EP_{pl} was 6.0 cm H₂O, and mean MP_{pl} was 3.2 cm H₂O.

Pleural Pressure as a Predictor of TP

In univariable analyses, greater IP_{pl} was associated with an increased risk of tension (OR, 1.40; 95% CI, 1.10-1.76). Similarly, greater EP_{pl} was associated with increased tension (OR, 1.22; 95% CI, 1.05-1.42). Greater IP_{pl} was associated significantly with increased risk of radiographic tension (OR, 1.31; 95% CI, 1.08-1.60) and was associated weakly with hemodynamically defined tension (OR, 1.30; 95% CI, 1.00-1.69). Greater EP_{pl} was found to be associated with increased risk of radiographic tension (OR, 1.19; 95% CI, 1.03-1.36), but not with hemodynamic tension (OR, 1.12; 95% CI, 0.95-1.31). These results are summarized in Table 2. A statistically significant association was found between MP_{pl} of > 0 cm H₂O and TP ($\chi^2 = 7.24$; $P = .007$). For the subtypes of TP, MP_{pl} of > 0 cm H₂O was associated with radiographic tension ($\chi^2 = 6.21$; $P = .013$), but not hemodynamic tension ($\chi^2 = 1.29$; $P = .257$); however, this study may be underpowered to detect a significant association.

Pleural Pressure as a Predictor of Clinical Outcomes

Lower IP_{pl} was associated with an increased risk of requiring pleurodesis (OR, 0.66; 95% CI, 0.45-0.96), but no association was found with in-hospital mortality (OR, 1.05; 95% CI, 0.92-1.20). No association was found between EP_{pl} and pleurodesis (0.88; 95% CI, 0.71-1.10).

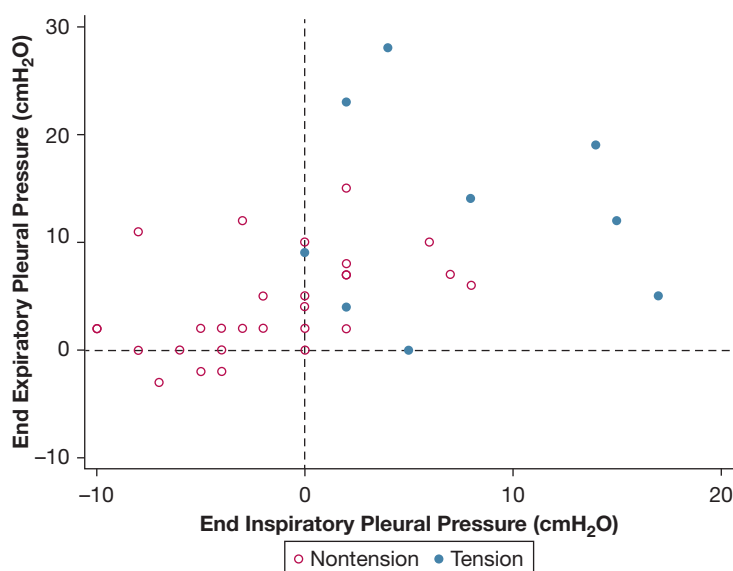


Figure 2 – Scatterplot showing observed end-inspiratory and end-expiratory pleural pressures for each study participant by tension status. Open circles represent patients without tension and closed circles represent patients with tension pneumothorax. Dashed lines denote atmospheric pressure.

or in-hospital mortality (OR, 1.00; 95% CI, 0.87-1.13). These results are summarized in [Table 3](#).

Sensitivity Analysis of Spontaneously Breathing Patients

Four patients were receiving mechanical ventilation at the time of pleural pressure measurement who were excluded from this sensitivity analysis. Notably, both patients with hemodynamic tension were receiving positive pressure ventilation. The results of this sensitivity analysis did not differ from those of the original analysis. Both IP_{pl} (OR, 1.45; 95% CI, 1.06-1.98) and EP_{pl} (OR, 1.18; 95% CI, 1.03-1.36) retained significant associations with tension. Lower IP_{pl} remained associated significantly with increased risk of pleurodesis (OR, 0.63; 95% CI, 0.42-0.95) and showed no association with in-hospital mortality (OR, 1.05; 95% CI, 0.87-1.27). EP_{pl} was not associated with need for pleurodesis (OR, 0.88; 95% CI, 0.70-1.11) or in-hospital mortality (OR, 1.01; 95% CI, 0.86-1.19).

Discussion

Our study is among the first to describe the direct measurement of pleural pressures using manometry

throughout the respiratory cycle in patients with pneumothorax. Normally, P_{pl} is between -3 and -5 cm H_2O at functional residual capacity while upright because of the combination of the tendency for the chest wall to expand and the elastic recoil forces of the lung. During normal spontaneous inspiration, pleural pressure becomes more negative as the diaphragm contracts to generate a gradient for airflow into the lungs. Expiration normally is a passive process in which the elastic recoil forces of the lung predominate and propel air out of the lungs.¹³ Therefore, in normal physiologic breathing, P_{pl} should remain subatmospheric throughout the respiratory cycle. However, in patients with active expiration (or labored breathing), respiratory muscles contract to generate muscular pressure, which augments total EP_{pl} and creates a gradient that forces expiration to occur. In this study, many patients exhibited this normal physiologic pattern ([Fig 2](#)). However, over the entire cohort, mean pleural pressures were positive, mean IP_{pl} was 0.5 cm H_2O , and mean EP_{pl} was 6.0 cm H_2O . The presence of positive IP_{pl} and EP_{pl} is important clinically because it directly increases the inspiratory and expiratory work required by the patient, which manifests as dyspnea and

TABLE 2] Associations Between Pleural Pressures and Subtypes of Tension Pneumothorax in the Study Cohort

Variable	Overall Tension (n = 9)	Radiographic Tension (n = 8)	Hemodynamic Tension (n = 2)
IP_{pl}	1.40 (1.10-1.76)	1.31 (1.08-1.60)	1.30 (1.00-1.69)
EP_{pl}	1.22 (1.05-1.42)	1.19 (1.03-1.36)	1.12 (0.95-1.31)

Data are presented as OR (95% CI). EP_{pl} = end-expiratory pleural pressure; IP_{pl} = end-inspiratory pleural pressure.

TABLE 3] Inspiratory and Expiratory Intrapleural Pressures as a Predictor of Clinical Outcomes

Variable	Pleurodesis	Death
IP _{pl}	0.66 (0.45-0.96)	1.05 (0.92-1.20)
EP _{pl}	0.88 (0.71-1.10)	1.00 (0.88-1.13)

Data are presented as OR (95% CI). EP_{pl} = end-expiratory pleural pressure; IP_{pl} = end-inspiratory pleural pressure.

respiratory distress. The measured values likely underestimate the component of IP_{pl} that is the result of air within the pleural space because of this additional component of muscular pressure.

The current definition of TP suggests that hemodynamic changes occur when P_{pl} persistently is positive and causes impairment in venous return, resulting in hemodynamic collapse. Our study is consistent with this definition because increased IP_{pl} and EP_{pl} both were associated significantly with the development of tension (OR, 1.40 and 1.22, respectively). Both increased IP_{pl} and EP_{pl} retained statistically significant associations with increased risk of radiographically defined tension (OR, 1.31 and 1.19, respectively). P_{pl} was not found to be statistically significantly associated with hemodynamically defined TP, but both values trended in the same pattern in this small sample size. Furthermore, the statistically significant association between MP_{pl} of > 0 cm H₂O and TP supports the notion that positive pleural pressures are responsible for the development of tension physiologic findings ($\chi^2 = 7.24$; $P = .007$).

The results of this study overall are consistent with the long-held belief that tension physiologic findings occur in the setting of supraatmospheric P_{pl} throughout the respiratory cycle. However, it is notable that 6 patients in the nontension cohort showed both supraatmospheric IP_{pl} and EP_{pl}. These findings suggest that supraatmospheric pleural pressures are necessary for tension to develop; however, elevated pleural pressures alone are insufficient for the development of TP. Bischin et al¹⁴ have suggested that additional elements beyond elevated pleural pressures are needed

for the development of hemodynamic collapse with TP and highlight the crucial roles of hypoxemia and mechanical ventilation. They propose that perfusion of the collapsed lung (physiologic shunt) causes hypoxemia that results directly in hemodynamic collapse, rather than true cardiogenic shock resulting from the elevated P_{pl} itself. However, this phenomenon occurs rarely in a spontaneously breathing patient with intact compensatory mechanisms. In mechanically ventilated patients, hemodynamic compromise occurs because of the significant increases in P_{pl} resulting from positive pressure ventilation as well as blunted compensatory mechanisms that augment venous return and maintain cardiac output. Consistent with these physiologic features, only 2 patients in this cohort demonstrated hemodynamic compromise, and both were receiving mechanical ventilation. This finding supports the role of mechanical ventilation in the presence of elevated P_{pl} for the development of hemodynamic collapse. Sensitivity analyses evaluating only spontaneously breathing patients did not alter the observed associations between pleural pressures and primary and secondary outcomes (Table 4). Ultimately, multiple factors contribute to tension physiologic findings and pleural manometry alone cannot predict it.

Analysis of secondary outcomes revealed no association of IP_{pl} (OR, 1.05; 95% CI, 0.92-1.20) or EP_{pl} (OR, 1.00; 95% CI, 0.88-1.13) with all-cause mortality during the same hospitalization. Given the small sample size, this study likely was underpowered to detect a true difference in mortality. Lower IP_{pl} was found to have a weak, but statistically significant, association with the need for pleurodesis (OR, 0.66; 95% CI, 0.45-0.96). Surprisingly, elevated IP_{pl} is protective against the need for future pleurodesis. One potential mechanism for this observation is related to the size of the pleural defect. In pneumothorax, air—and therefore pressure—accumulates in the pleural space because of a 1-way ball-valve defect in the pleura. Smaller defects may trap air better in the pleural space, resulting in greater P_{pl}, but therefore also may heal more easily. However, larger pleural defects may allow more air to escape the pleural space, resulting in overall lower P_{pl}, but may not heal as

TABLE 4] Sensitivity Analysis of Association Between Pleural Pressure and Clinical Outcomes in Spontaneously Breathing Patients

Variable	Tension	Pleurodesis	Mortality
IP _{pl}	1.45 (1.06-1.98)	0.63 (0.42-0.95)	1.05 (0.87-1.27)
EP _{pl}	1.18 (1.03-1.36)	0.88 (0.70-1.11)	1.01 (0.86-1.19)

Data are presented as OR (95% CI). EP_{pl} = end-expiratory pleural pressure; IP_{pl} = end-inspiratory pleural pressure.

easily and therefore would require further intervention. However, this ball-valve mechanism has not been validated and requires further investigation.

Our study does have some limitations. Participant recruitment occurred over 34 months at 5 different sites, but only 37 patients were enrolled. Recruitment was performed passively because it relied on referral to the study team, who largely were pulmonologists. Much of the study occurred during peak times of the COVID-19 pandemic, also limiting recruitment. Most participants were enrolled after other clinically indicated procedures were performed by members of the study team, which explains the high proportion of patients with iatrogenic pneumothorax and lower proportion of patients with traumatic pneumothorax enrolled, which may limit generalizability to other pneumothorax causes. Also, most patients in this study had known underlying structural lung disease that might have influenced pulmonary elastance, and therefore might have affected pleural pressures, so the results of our study should be interpreted carefully in this context. Furthermore, we enrolled 9 patients with TP and 5 patients died during hospitalization, which highlights that our study population was particularly ill. The digital manometer limited the study to pressure readings at end inspiration and end expiration; therefore, our study does not account for pleural pressure during the rest of the respiratory cycle. The

overall small sample of this study further limits this study's power and the conclusions that can be drawn from these data. Follow-up studies with additional participants are needed to validate our findings.

Interpretation

This is among the first studies to report direct measurements of pleural pressures in patients with clinical pneumothoraxes. The findings of supraatmospheric IP_{pl} and EP_{pl} in the tension cohort support the long-held definition of TP. Because multiple patients in the nontension cohort displayed persistently supraatmospheric IP_{pl} and EP_{pl} , this suggests that other factors outside of increased pleural pressures alone are needed for tension to develop. The association of IP_{pl} with the need for pleurodesis shows that pleural manometry may have implications for the management of pneumothorax. Additional larger prospective studies are needed to validate these findings.

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